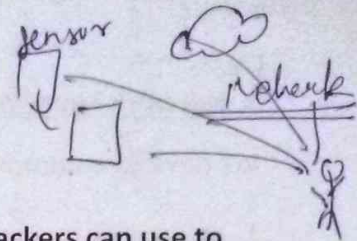


Vulnerabilities of IoT

Vulnerability -

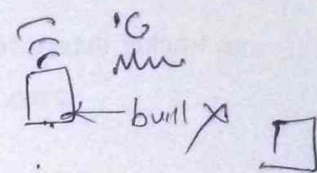
A vulnerability is a weakness or flaw in a system that attackers can use to hack or steal data from it.



1. Sensor-Level Vulnerabilities

Sensors collect data (temperature, motion, heart rate, etc.).

They are often small,
no built-in security.



How to Handle

Use secure boot

Regular device inspection

Ex:

A smart temperature sensor in a factory is physically accessed and altered to show false readings.

2. Device-Level Vulnerability.

A device-level vulnerability is a weakness inside the IoT device itself that allows attackers to steal data.



How to Handle:

Secure boot

Trusted hardware modules

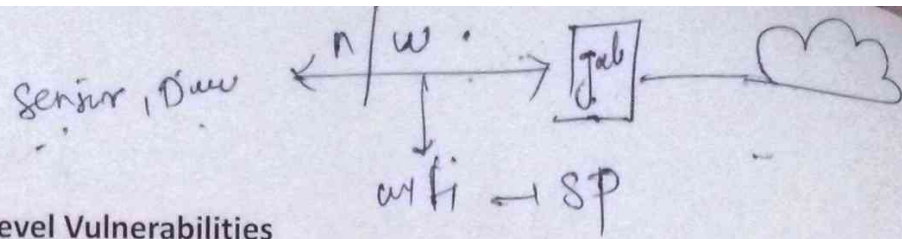
Ex:

A smart CCTV camera uses:

Username: admin

Password: admin

Attackers try common passwords and gain full access.



3. Network-Level Vulnerabilities

IoT devices communicate over networks (Wi-Fi, Bluetooth, Zigbee).

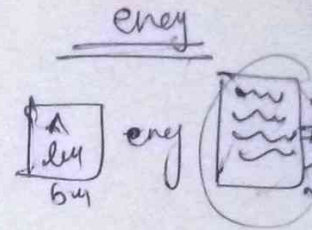
How to Handle

Secure Wi-Fi with strong passwords

VPNs

Ex:

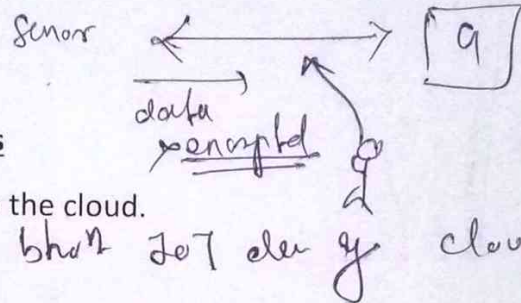
A hacker listens to unencrypted data sent from a smart meter.



4. Gateway-Level Vulnerabilities

Gateways connect IoT devices to the cloud.

→ act as middle man b/w IoT devices & cloud.



How to Handle

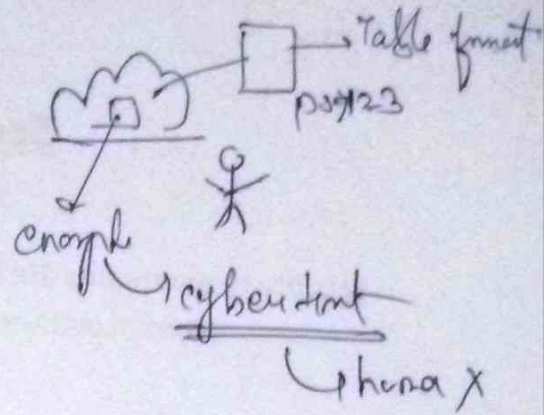
Regular gateway inspection Use intrusion detection systems (ALARM)

Ex:

If the gateway is hacked, all connected devices are hacked.

5. Cloud Vulnerabilities

IoT data is stored and processed in the cloud.



How to Handle

Encrypt data

Regular security

Ex:

User data stored in the cloud is leaked due to open database access.

6. User-Level Vulnerabilities

Humans are often the weakest link.

Vulnerability of IoT

✓ Weak passwords

✓ Lack of security awareness

✓ Reusing passwords

How to Handle

User education & awareness

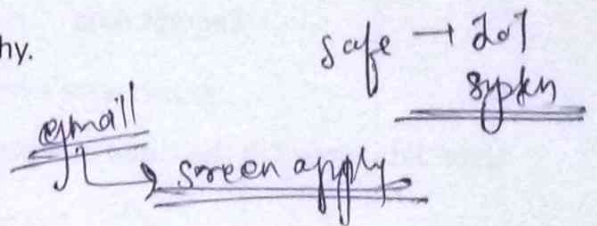
Strong password

Ex:

weak password → the great
gray

Security Requirements

Security requirements are the basic protections needed to keep an IoT system
safe
and
trustworthy.

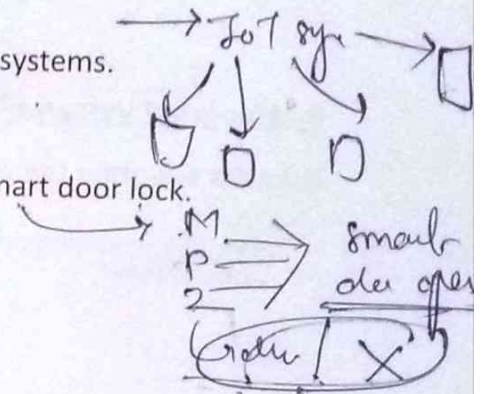


1. Authentication

Authentication ensures that only authorized devices and users can access the IoT system.

Strong authentication increases trust and security in IoT systems.

Example: Only a registered mobile phone can unlock a smart door lock.



2. Authorization (Access Control)

① Authorization defines what actions a user or device is allowed to perform after authentication.

Example: In a smart home, guests can control lights but cannot access security cameras.

② There are list of thing which users can do and cannot do in IoT/application.

Swiggy
login

do
- order
- add change
- line out.

can't do
- listing out of restaurants

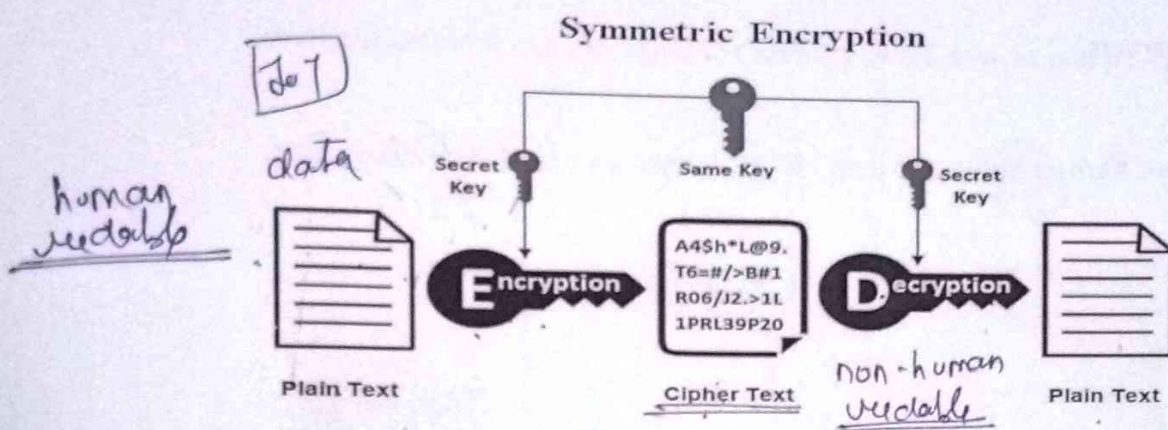
3. Data security:

It ensures that IoT data remains private and secret.

It prevents unauthorized people from

reading sensitive information.

Security is maintained using encryption techniques.



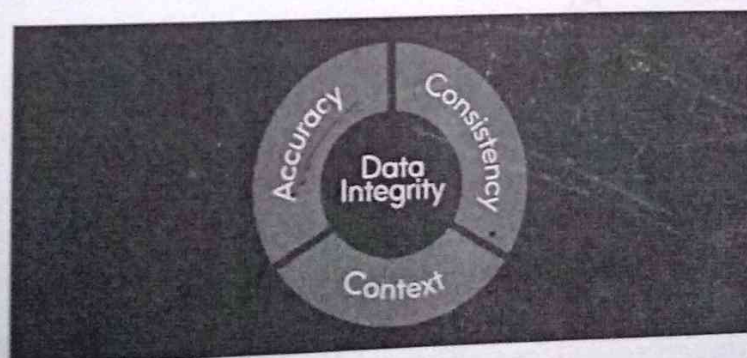
Example: Data sent from a wearable health device is encrypted so hackers cannot read it.

4. Data Integrity

Data integrity ensures that data is not changed during transmission or storage.

Modified or fake data can lead to wrong decisions or system failure.

Example: Smart meter readings reach the server without being changed by attackers.



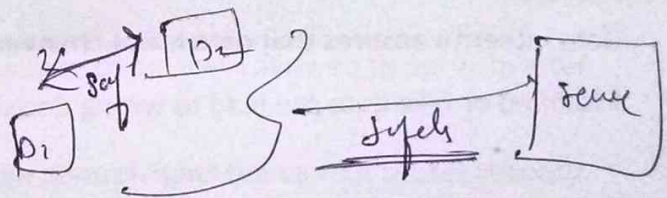
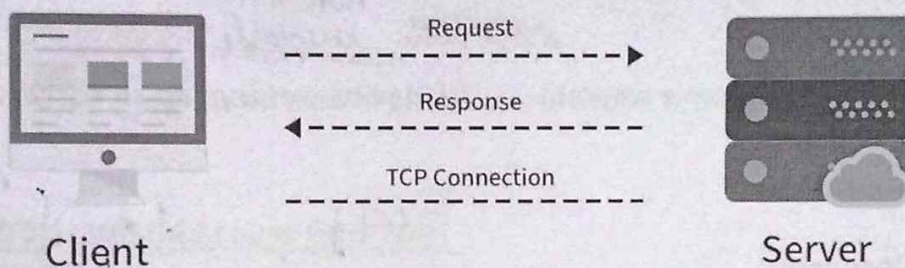
Happiness
Happiness

5. Secure Communication

Secure communication protects data while it travels between devices and cloud servers.

Example: A smart home app uses HTTPS for secure device communication.

HTTP Connection

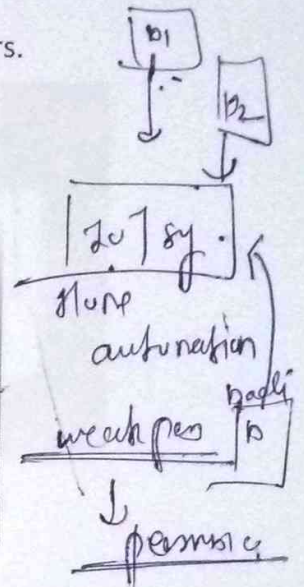


Key to
Server to
Challenges
Server to

Challenges for Secure IoT

1. Authentication

Authentication is used to verify the identity of IoT devices and users.



Many IoT devices use weak or default passwords.

This allows fake or malicious devices to enter the network.

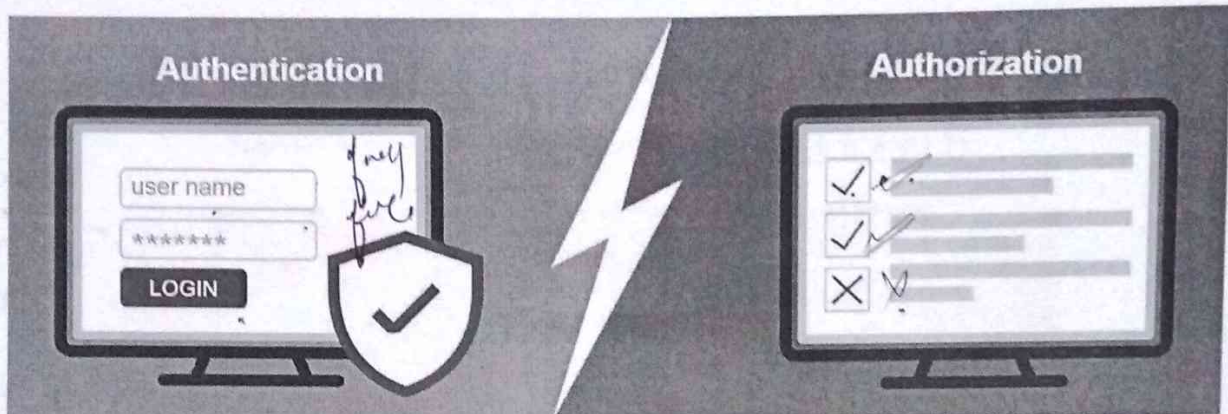
Example: A smart camera gets hacked because it uses a default username and password.

u:- admin
p:- admin

2. Authorization.

(Access control)

Authorization controls what actions a user or device is allowed to perform



Poor access control allows users to access unauthorized functions.

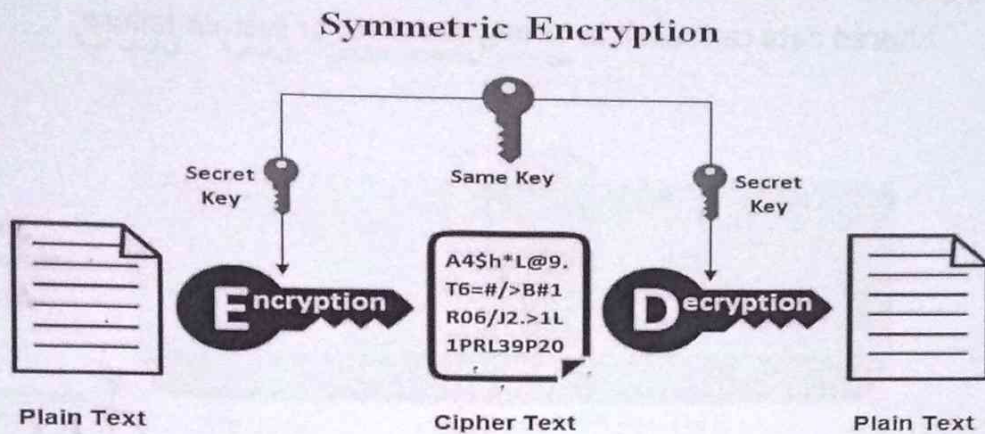
Lack of role-based permissions increases misuse of devices.

Pen gc

Example: A guest user changes security settings in a smart home system.

3. Data Security (Confidentiality)

Data security ensures that IoT data remains private and protected.

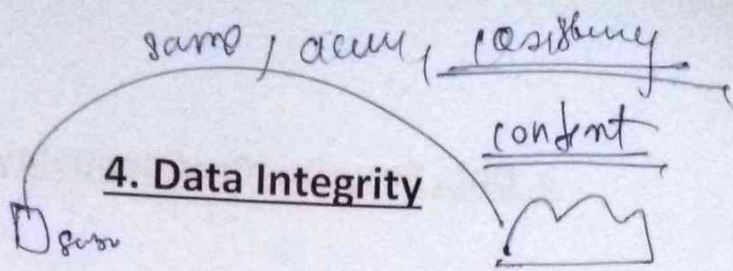


Data is often transmitted without encryption.

Attackers can easily read sensitive information.

This causes privacy and data leakage issues.

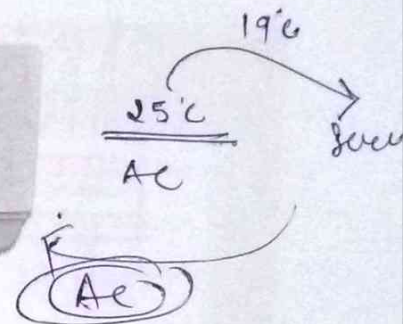
Example: Health data from a wearable device is intercepted on an open Wi-Fi network.



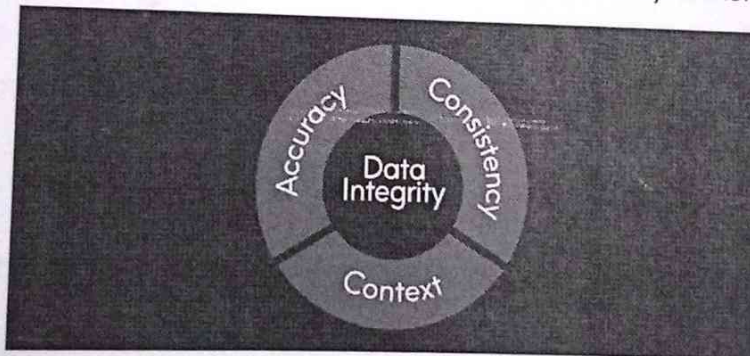
Data integrity ensures that data is not modified during transmission or storage.

Attackers may change data to mislead the system.

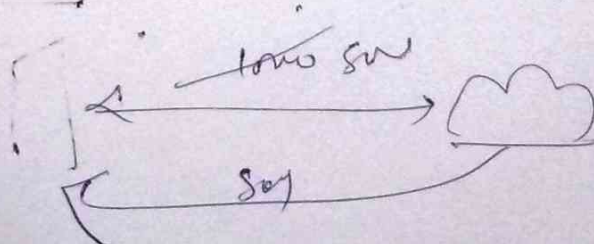
Altered data can result in wrong decisions or system failure.



Detecting data modification is difficult in IoT systems.

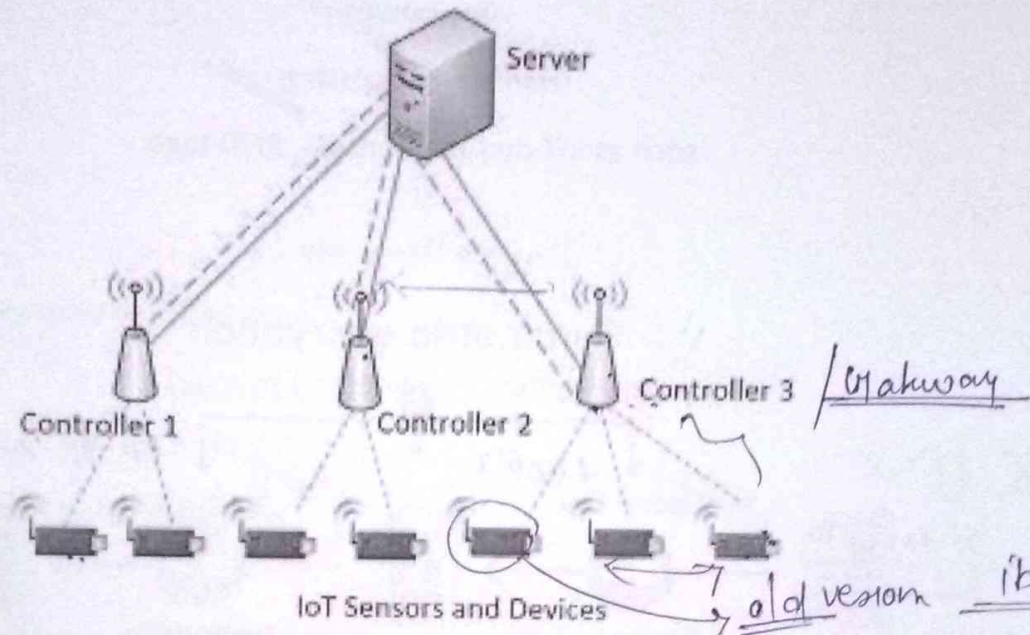


Example: Smart meter readings are altered to reduce electricity bills.



5. Secure Communication

Secure communication protects data exchange between IoT devices and servers.



Many IoT systems use insecure or outdated communication protocols.

Data can be intercepted or altered during transmission.

This leads to unauthorized control of devices.

Example: A hacker sends fake commands to a smart bulb over an unsecured network.

⑤ choice of
secured
IoT

Lightweight Cryptography

Lightweight cryptography refers to encryption techniques specially designed for devices with

limited resources

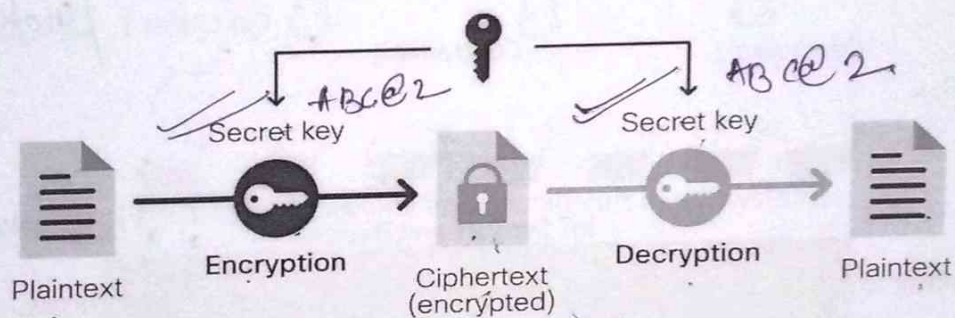
less memory

low power

minimal processing

such as IoT devices, sensors, RFID tags.

Symmetric encryption



1. Lightweight cryptography is used in environments where hardware resources like CPU, memory, and battery are limited.
2. It provides basic security features such as integrity, and authentication with low computational cost.

3. Algorithms are designed to be

Simple ✓

Fast ✓

and

energy-efficient ✓

4. It is mainly used in IoT systems.

Types of Lightweight Cryptography

1. Symmetric Key Cryptography

1. Uses the **same secret key** for both encryption and decryption.

2. Requires

less computation ✓

less memory ✓

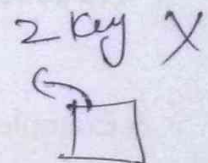
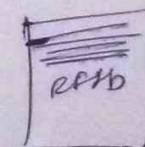
making it suitable for IoT devices. ✓✓

3. Faster than public key cryptography.

4. Example:

AES (lightweight versions),

used in RFID systems.



Ma

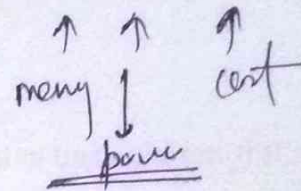
eng

dec

2. Public Key Cryptography

1. Uses two keys: a public key for encryption
and
a private key for decryption.

2. Provides better authentication.



3. Requires more computation than symmetric methods.
4. Requires more memory than symmetric methods.

5. Example:

Elliptic Curve Cryptography (ECC) used in IoT applications.

signature (A)



Threat Model in IoT Security

any
in
→ policy

A threat model is a method to identify possible

attacks, attackers

and weak points in an IoT system

Threat Model for IoT Applications

1. Assets

Assets ===== valuable things
in an IoT system that attackers want.

✓ IoT devices (sensors, cameras, meters)

✓ Data (personal, medical, industrial)

Network connections

Cloud storage and servers

Example:

Health data collected by a smart wearable is an important asset.



2. Threat Actors

Threat actors are people or systems that perform attacks.

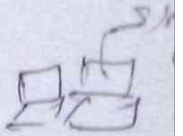


Hackers

Cyber criminals

Insider attackers (employees)

Unauthorized users



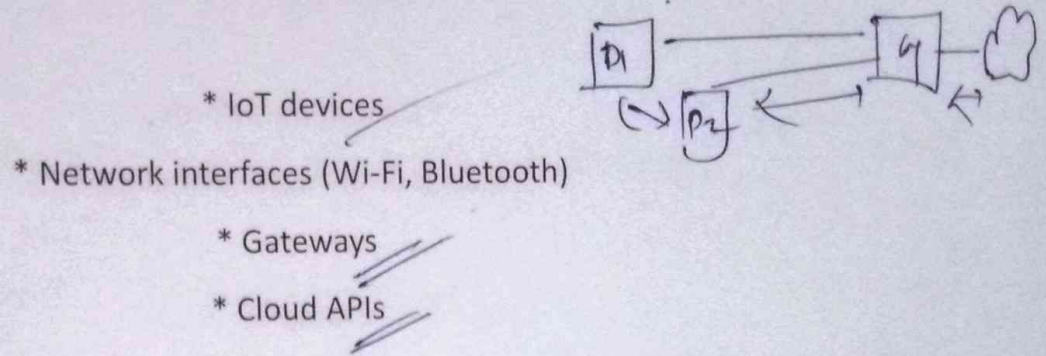
Example:

A hacker tries to access a smart home camera from the internet.

3. Entry Points (Where attack starts?)

wifi
weak
unsec

Entry points are places where attackers can enter the system



Example:

An attacker enters through an unsecured Wi-Fi connection of a smart device.

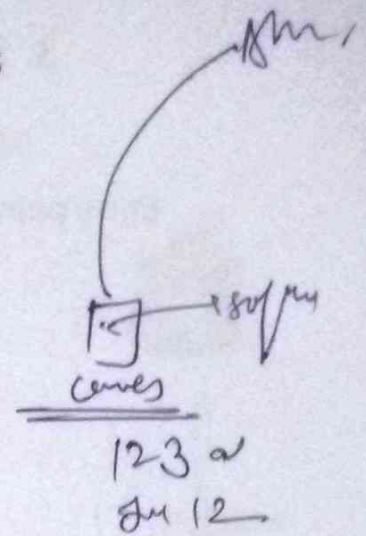
4. Threats at Different IoT Layers

a) Device-Level Threats

- * Default passwords.
- * Malicious firmware installation

Example:

Attacker resets a smart camera and installs fake firmware.

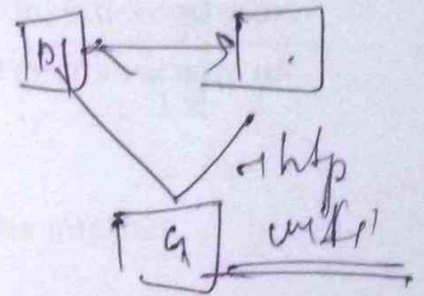


b) Network-Level Threats

- * Man-in-the-middle attack
- * Denial of Service (DoS)

Example:

Attacker intercepts data sent from sensor to gateway.

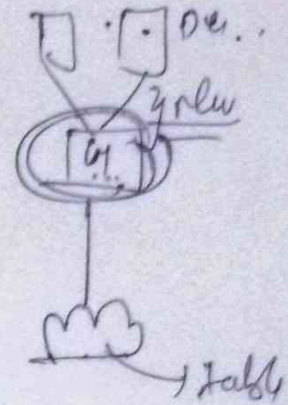


c) Gateway-Level Threats

- * Weak authentication
- * Single point of failure

Example:

Hacking the gateway gives access to all connected devices.

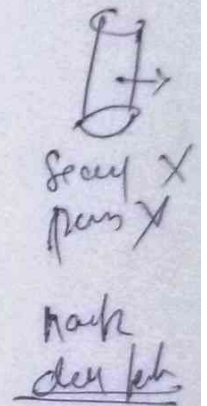


d) Cloud Level Threats

- * Data breaches
- * Misconfigured cloud storage

Example:

User data leaked due to open cloud database.

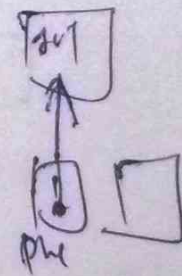


e) Application / User-Level Threats

- Weak passwords
- Insecure mobile apps

Example:

User clicks a fake link and attacker controls IoT devices.



6. Impact of Threats.

Data loss

1. Privacy loss

2. Financial loss

3. Safety risks

4. Loss of user trust

Breach



7. Security Controls (How threats are reduced?)

* Authentication and authorization

* Encryption

data secure

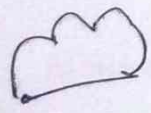
* Monitoring and intrusion detection

data integrity

control

P1

G



probe

Alarm or

Alert
msg